Mahagama Ralalage Dilusha Heshan Hemachandra

Mechanical Engineering Undergraduate

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Profile / Summary

Strongly interested in Robotics, Mechatronics, and Industrial Automation, with a passion for designing intelligent robotic systems and autonomous solutions.

Hands-on experience in CAD design, embedded systems, PLC programming, PCB design, Python/C++ programming, and machine learning through coursework, individual and group projects.

Seeking an internship in **Automation or Research & Development (R&D)** to apply practical engineering skills and gain industry experience.

Education

University of Moratuwa, Sri Lanka — **2024 – Present**
B.Sc. Engineering in Mechanical Engineering — Semester 5
CGPA: 3.09/4.00

Pinnawala Central College, Sri Lanka — **2020 – 2023**
G.C.E. Advanced Level, Physical Science Stream – Combined Mathematics, Physics (A), Chemistry (A)
Z-Score: 2.2548 | Island Rank: 532

G.C.E. Ordinary Level Examination
8 A passes and 1 B pass

Technical Skills

Mechanical Design	SolidWorks, AutoCAD
Simulation & Analysis	MATLAB, Webots, ANSYS (Basic FEA & CFD)
Programming & ML	Python, C++, Machine Learning
PCB Design	Altium Designer
Embedded & Robotics	ESP32, LoRa, Raspberry Pi, Arduino
Industrial Automation	Basic PLC Programming
Manufacturing	Rapid Prototyping, Design for Manufacture
Languages	Sinhala (Native), English (Working Proficiency)

Selected Projects

Pattiya – Smart IoT Cattle Collar — **2026 – Present**
Team Innovation Project | AgriTech, IoT and Product Development — *Semi-Finalist – SPARK Challenge 2026*

- Developing an integrated cattle collar for real-time methane, estrus, heat-stress and geofence/security monitoring using gas, temperature, motion, GPS and acoustic sensing.
- Developing machine-learning models to analyse multi-sensor data and identify patterns related to estrus, heat stress and cattle health monitoring.
- Contributed to a weather-resistant CAD enclosure and internal component layout suitable for a wearable device operating in farm environments.
- Supported ESP32 sensor integration and LoRa/LoRaWAN communication between the collar, environmental station and farmer-facing alert system.

Tech Stack: SolidWorks, Altium Designer, ESP32, LoRa, Raspberry Pi, Python, Machine Learning

RoboRoarZ PathFinder – Autonomous Maze-Solving Robot — **2026**
Team Competition Project — *Champion – RoboRoarZ Sri Lanka 2026*

- Co-developed an autonomous differential-drive robot capable of detecting maze walls, exploring routes and executing a shortest-path run under competition constraints.
- Contributed to chassis integration, sensor placement, motor-driver connections and embedded control implementation.
- Tested and tuned turning, wall detection and cell-to-cell movement for reliable competition performance, contributing to the championship result.

Tech Stack: C++, Arduino, SolidWorks

Selected Projects (Continued)

Reverse Engineering of a Motorcycle Helmet – EPS Liner Analysis **2026**
Academic Design Project *Mechanical Design and Product Analysis*

- Disassembled and dimensionally inspected a motorcycle helmet to understand the relationship between the outer shell, comfort padding and expanded-polystyrene impact liner.
- Reconstructed key helmet and EPS-liner geometry as detailed CAD models using measured dimensions and sectional features.
- Examined liner thickness, segmentation and energy-management features, documenting observations related to impact protection and manufacturability.

Tech Stack: SolidWorks, AutoCAD, Reverse Engineering, Dimensional Metrology

Team Bumblebee – SLRC 2026 Autonomous Dual-System Robot **2026**
Team Competition Project | Simulation and System Integration *Finalist – Sri Lankan Robotics Challenge 2026*

- Collaborated on a dual physical/virtual autonomous system designed to operate under real-time competition constraints and changing obstacle conditions.
- Supported robot simulation and integration of the physical platform with the Ares virtual robot.
- Contributed to an API-based communication workflow that transferred decoded coordinates and commands for parallel navigation and dynamic obstacle handling.

Tech Stack: Python, Raspberry Pi, Arduino, Webots

Micromouse PathFinder Robot **2025**
Team Competition Project *Finalist – SLIIT ROBOFEST 2025*

- Designed and built an autonomous micromouse platform for maze exploration, wall detection and shortest-path solving.
- Integrated distance sensors, motor drivers and embedded control electronics within a compact mobile-robot chassis.
- Implemented flood-fill path-planning logic and tuned motion for reliable straight-line tracking, repeatable turns and route optimisation.

Tech Stack: C++, Arduino, SolidWorks, Altium Designer

Smart Wall Painting Robot – Developed an automated robotic system for wall-painting applications, combining mechanical design, motion control and hardware–software integration to improve painting consistency and reduce manual effort. The project provided hands-on experience in robot fabrication, function development, programming and embedded hardware integration.
Tech Stack: SolidWorks, Arduino, C++

Ongoing Projects

Fish Feed Extruder Gearbox Design – designing and developing a multi-speed gearbox with forward and reverse gear arrangements to provide suitable speed and torque for fish-feed extrusion, while enabling controlled screw reversal for blockage recovery and maintenance.

UAV Self-Leveling Landing Gear – designed and optimized a multi-link passive landing gear mechanism to maintain stable, near-vertical foot motion and adapt to uneven terrain during landing.

Achievements

- **Champion** – RoboRoarZ Sri Lanka International Robotics Competition, Team PathFinders, 2026
- **Finalist** – Sri Lankan Robotics Challenge (SLRC), Team Bumblebee, 2026
- **Semi-Finalist** – SPARK Challenge 2026, Team Pattiya, University of Moratuwa
- **Finalist** – SLIIT ROBOFEST 2025, University Category – Micromouse

Reference

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